

Cognitive Neuroscience Term Project

Report 4: Data Analysis

Team Axion

1) Data processing

1.1 Excluded data

A total of **20 participant CSV files** were collected. Two files/participants were excluded prior to analysis based on predefined criteria: **one dataset was a pilot/test run** (the file containing “P18”) and was therefore not part of the experimental sample, and **one participant was excluded due to age** (P04, 14 years old, below the minimum age criterion). After these exclusion **final sample size was N =18 participats** which is also confirmed by the statistical analysis dataset (**18 unique participant**).

Of the 18 participants in datasets 10 are women and 8 are men. This is considered a balanced distribution. Sixteen of these participants are right handed, while only two are left handed. Since this distribution is not balanced, we did not include right handed/left handedness in the analysis.

1.2 Data description

Raw data were recorded separately for each participant as **PsychoPy exported CSV files** (one file per participant/session). Each file follows a **trialwise structure: rows correspond to individual trials and columns contain stimulus descriptors, responses, and timing informatin**. In a typical raw file, there is an initial row containing session-level information, followed by **120 trial rows**.

For each trial, the raw dataset includes (i) the presented **stimulus file path** (e.g., stimuli/F15NE.jpg), (ii) the experimental condition labels, including **face gender** (Gender: Male/Female) and **facial emotion** (Emotion: Neutral/Happy/Angry), (iii) the **expected response** (CorrectAns: f or m), (iv) the participant’s **key press** (key_resp.keys), (v) **accuracy** (key_resp.corr, coded 1/0), and (vi) **reaction time** (key_resp.rt, in seconds). The raw export also contains numerous PsychoPy bookkeeping and timing fields (e.g., trial indices and timestamps such as thisTrialN, thisN, trial.started/stopped, image.started, etc.), which are not required for the planned behavioral analyses.

Participant-level metadata are stored in the same CSV as repeated columns across trials, including **participant identifier, session, age, participant gender** (notably recorded as gender), **education years, handedness**, and experiment/runtime information (e.g., date/time stamp, PsychoPy version, frame rate). Because the raw export uses both Gender (stimulus face gender) and gender (participant gender), this naming ambiguity was handled later during

data preparation/analysis via explicit renaming to keep stimulus and participant variables distinct.

1.3 Preprocessing of data from each participant

Each participant's PsychoPy CSV file was processed using a standardized pipeline before inclusion in the combined dataset. First, files were read with **UTF-8-SIG** encoding to prevent character/formatting issues. Nontrial rows were removed by dropping rows where the **stimulus identifier**(ImageFile) was missing; this step ensures that only actual experimental trials (i.e., trials with a presented face stimulus) are retained. For consistency across participants only the analysis-relevant columns were kept: **participant metadata** (participant, age, handedness, gender) and **trial level task variables** (stimulus face gender Gender, emotion Emotion, reaction time key_resp.rt, and accuracy key_resp.corr). Any trial with missing values in these critical fields was excluded from further analysis.

After merging the participant files into a single dataset, reaction-time quality control was applied in two stages.

1. **Global RT trimming:** trial with RTs outside **0.2–3.0 seconds** were removed (RT < 0.2s typically reflects anticipatory/accidental responses; RT > 3.0 s is interpreted as attentional lapse). This step removed **61 trials (2.82%)**, leaving **2099 trials**.
2. **Within subject outlier removal:** to control for individual differences in overall speed, RT outliers were removed using **z scores within each participant and condition cell** (grouped by participant × Gender × Emotion). Trials with $|z| > 2.5$ were excluded. This removed **72 additional trial (3.43% of the trimmed data)**, yielding **2027 trials**.

Overall, **6.16% of trials** were excluded through the combined missing-data filtering, global trimming, and within-subject outlier removal steps.

2) Subject based analysis

Subject based (participant level) summary measures were computed after trial level cleaning to prepare the dataset for inferential statistics. For each participant, **conditionwise aggregates** were generated for the **2 (face gender: Female, Male) × 3 (emotion: Angry, Neutral, Happy)** factorial design yielding **6 summary rows per participant**. With **N = 18 participants**, this resulted in **108-row subject summary table** (18 × 6), which was saved as `final_summary_for_anova.csv`.

Reaction time (RT) computation

Reaction time was summarized **only using correct trials** to avoid contaminating speed estimates with guesses or incorrect key presses. Within each participant and condition cell (**participant × face gender × emotion**), RT was computed as the **mean of key_resp.rt (seconds)** over trials where `key_resp.corr == 1`. The resulting variable was stored as `mean_rt`.

Accuracy computation

Accuracy was computed using **all valid trials** (after NaN removal and RT trimming/outlier steps). Within each participant and condition cell (**participant** × **face gender** × **emotion**), accuracy was calculated as the **mean of key_resp.corr** (coded 1 = correct, 0 = incorrect). This yields a proportion in ([0, 1]) and was stored as `accuracy`.

Example (single participant; 6 condition cells)

Below is an illustrative example of the subject-level output structure (participant “P01”), showing the six cells produced by the aggregation:

participant	face gender	emotion	accuracy	mean_rt (s)
P01	Female	Angry	0.8421	0.7499
P01	Female	Happy	0.9500	0.7646
P01	Female	Neutral	0.8947	0.7841
P01	Male	Angry	0.9000	0.7782
P01	Male	Happy	1.0000	0.8377
P01	Male	Neutral	1.0000	0.7696

In addition to these outcome variables, participant-level metadata (e.g., age, handedness, and participant gender—kept distinct from stimulus face gender) were merged into the same summary table to allow reporting of sample characteristics alongside the condition-wise behavioral measures.

3) Preparation for statistical analysis

All inferential analyses were conducted on a **participant-level summary dataset** created after trial-wise cleaning and aggregation. Specifically the file `processed_data/final_summary_for_anova.csv` contains one row per **participant** × **face gender** × **emotion** condition cell ($2 \times 3 = 6$ cells per participant). The table include dependent variables **mean RT** (`mean_rt`, seconds; computed on correct trials) and **accuracy** (`accuracy`, proportion correct), along with participant descriptors (**participant gender**, **age**, **handedness**)

Before running statistical tests, the dataset was prepared as follows:

1. **Variable harmonization and naming clarity.**

To prevent ambiguity between participant gender and stimulus face gender column were explicitly renamed so that Gender refers to **face gender** (`face_gender`) and gender refers **participant gender** (`participant_gender`).

2. **Factor coding and ordering.**

The within-subject factors were treated as categorical variables with a fixed level order for consistent reporting and plotting:

- Emotion: **Angry** → **Neutral** → **Happy**
 - face_gender: **Female** → **Male**
3. **Completeness and integrity checks.**
The summary dataset was checked to ensure that all condition cell required for repeated measures testing were present. No missing cells were detected for either dependent variable (**Missing RT cells = 0; Missing accuracy cells = 0**) and the final dataset contained **18 unique participants** (i.e., $18 \times 6 = 108$ condition row)
 4. **Distributional checks (normality).**
Given the relatively small sample size, normality was assessed using a **Shapiro–Wilk test** on **participant-level means** (rather than cell-level values, which can be noisy with few trials per cell). This check indicated that **RT participant means deviated from normality** ($p = 0.0113$), whereas **accuracy participant means were compatible with normality** ($p = 0.3414$). These diagnostics were used to guide interpretation; however, the primary analysis proceeded with the RM-ANOVA approach required by the study design.
 5. **Sphericity handling and reporting convention.**
For repeated-measures effects involving a factor with more than two levels (i.e., **Emotion** and the **face_gender × Emotion** interaction), **Greenhouse–Geisser (GG) correction** was applied and **GG-corrected p-values** were reported together with the **GG epsilon**. Sphericity is not applicable to the two-level factor (**face_gender**).

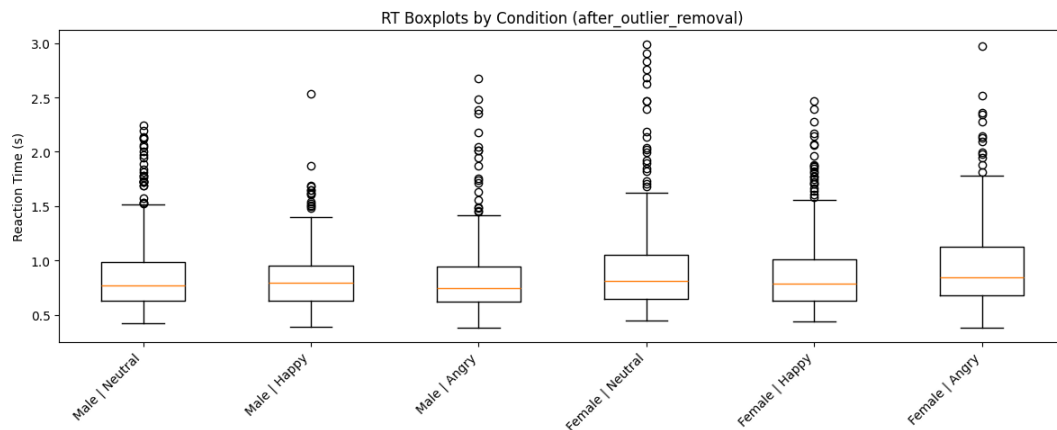
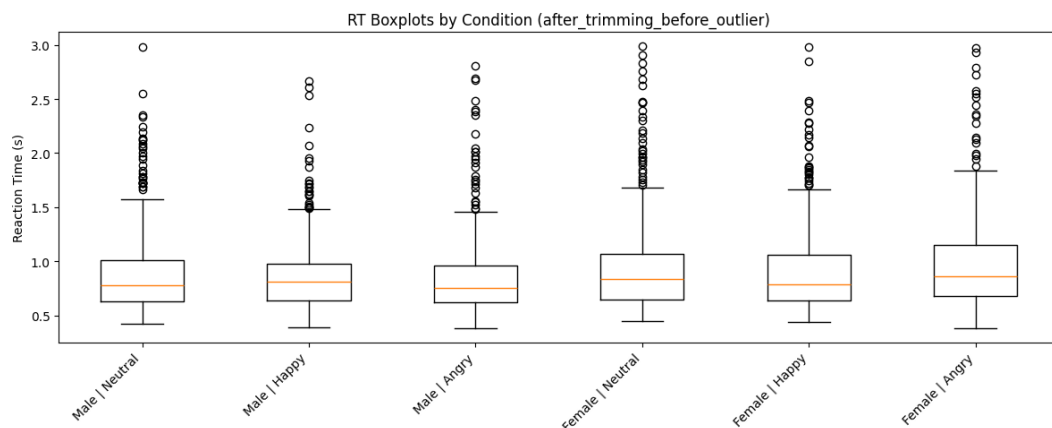
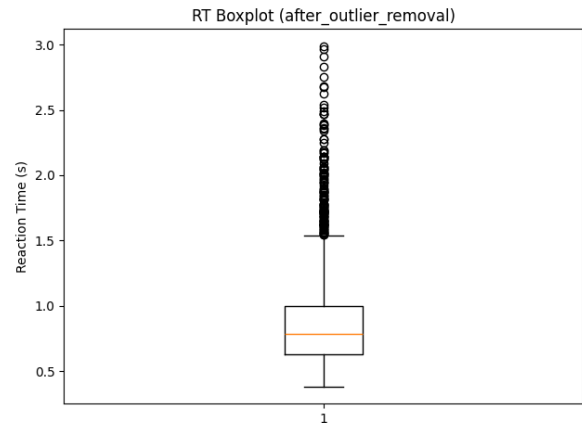
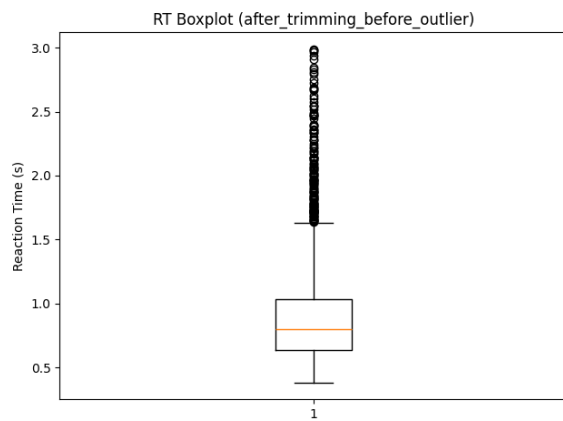
All main-model outputs (RM-ANOVA tables for RT and accuracy) were exported as CSV files to `results/main_analysis/`, and short report-ready result sentences were automatically generated and saved as `results/main_analysis/report_snippets.txt`.

4) Box Plots / Bar Plots (Visualization)

To document data quality and support the preprocessing decisions, reaction-time (RT) distributions were visualized using box plots both **before** and **after** within-subject outlier removal. These plots serve two purposes: (i) to confirm that RTs show the expected **right-skewed distribution** typical for speeded decision tasks, and (ii) to make the impact of the cleaning steps transparent by showing how extreme values are reduced while the central tendency and interquartile ranges remain stable.

First, RT box plots were generated for each of the **six experimental cells** (face gender × emotion) after global trimming (0.2–3.0 s) but **before** z-score-based outlier removal. This condition-wise visualization highlights between-condition similarities and differences in median RT and spread, and it also makes it clear that a small subset of trials still exhibits unusually long RTs. In parallel, an overall RT box plot (collapsing across conditions) provides a compact summary of the full trial-wise distribution and the prevalence of upper-tail values.

Next, the same visualizations were produced **after outlier removal** ($|z| > 2.5$ within each participant × face gender × emotion cell). Comparing the “before” vs. “after” plots confirms that the procedure primarily attenuates the **upper tail** (very long RT trials) without materially distorting the **median** and **IQR**, supporting the interpretation that the outlier step removes lapses rather than shifting the core response-speed distribution.



5) Data Analysis

5.1 Statistical tests

All hypothesis testing was conducted on the **subject-level summary dataset** (one row per **participant** × **face gender** × **emotion** cell; 6 rows per participant). The study design is a fully within-subject **2 × 3 repeated-measures factorial design**, with the within-subject factors:

- **Face gender** (2 levels: *Female*, *Male*)
- **Emotion** (3 levels: *Angry*, *Neutral*, *Happy*)

Two dependent variables(DVs) are analyzed separately:

- **Reaction time (RT):** `mean_rt` (seconds; computed from correct trials only)
- **Accuracy:** `accuracy` (proportion correct)

Assumption checks

Given the small sample size ($N = 18$), normality was assessed using the **Shapiro–Wilk test** on **participant-level means** (collapsed across conditions) for each DV. This check suggested a deviation from normality for RT participant means, while accuracy participant means were consistent with normality. These diagnostics were used to guide interpretation, while the prerequired main analysis approach (RM-ANOVA) was retained.

Main inferential model: 2×3 RM-ANOVA

For each DV a **twoway repeated measures ANOVA** was performed with `participant` as the subject identifier and `face_gender` and `Emotion` as within-subject factors. The analysis was implemented using a detailed RMA NOVA output (reporting sums of squares, degrees of freedom, mean squares, **F-statistics**, and effect size).

Because sphericity can be violated for effects involving **Emotion** (3 levels) and the **face_gender $\times$ Emotion** interaction, **Greenhouse–Geisser (GG) correction** was applied where applicable, and **GG-corrected p-values** (`p-GG-corr`) were reported alongside the **GG epsilon (ϵ)**. For the two level factor **face_gender**, sphericity is not applicable and the uncorrected p-value is equivalent to the corrected value.

All statistical tests used a two sided significance threshold of $\alpha = 0.05$.

Post-hoc testing strategy

Posthoc comparisons were conducted **only when the omnibus RM ANOVA effect was significant**:

- If the **face_gender $\times$ Emotion interaction** was significant the interaction was decomposed via **simple effects analyses**: within each facegender level (*Female* and *Male* separately), **pairwise comparisons across Emotion levels** were performed using **paired-samples tests**.
- If the interaction was not significant but the **main effect of Emotion** was significant, pairwise comparisons across Emotion levels were performed on the full dataset.

To control family-wise error due to multiple testing, p-values for pairwise comparisons were adjusted using the **Holm correction** (`padjust='holm'`).

5.2 Results

5.2.1 Reaction Time statistics

Reaction time (RT) was analysed using a **2(face gender: Female, Male) × 3 (emotion: Angry, Neutral, Happy) repeated-measures ANOVA** on the participant level summary dataset (N = 18; 6 condition means per participant). RT values represent **mean correct trial RTs** in seconds after trial-level trimming (0.2–3.0 s) and within-subject outlier exclusion ($|z| > 2.5$ within *participant × face gender × emotion*).

RM-ANOVA results (RT, mean_rt).

- **Main effect of face gender:** $F(1, 17) = 4.437$, $p = 0.0503$, $\eta^2 = 0.0326$. This effect was **not statistically significant** at $\alpha = .05$, although it was close to the threshold.
- **Main effect of emotion:** $F(2, 34) = 2.139$, $p_{GG} = 0.1449$, $\varepsilon_{GG} = 0.8053$, $\eta^2 = 0.00637$ (not significant)
- **Face gender × emotion interaction:** $F(2, 34) = 1.245$, $p_{GG} = 0.2966$, $\varepsilon_{GG} = 0.8135$, $\eta^2 = 0.00343$ (not significant)

Given the absence of a significant omnibus effect for **Emotion** and the **interaction**, no emotion level posthoc comparisons were conducted for RT.

Descriptive pattern (means ± SD across participants).

Condition means suggested generally slower responses for **Female face** stimuli than **Male face** stimuli across emotions:

- Female face Angry **0.9709 ± 0.2911** , Neutral **0.9484 ± 0.3053** , Happy **0.9079 ± 0.2719**
- Male faces: Angry **0.8423 ± 0.2022** , Neutral **0.8841 ± 0.2438** , Happy **0.8344 ± 0.1399**

This descriptive trend is consistent with the near threshold face gender effect but it should be interpreted cautiously given the non significant omnibus result.

5.2.2 Accuracy statistics

Accuracy was analysed using the same **2(face gender: Female, Male) × 3 (emotion: Angry, Neutral, Happy) repeated measures ANOVA** on participant level summary dataset (N = 18; 6 condition means per participant). Accuracy values represent **proportion correct** after trial level cleaning (missing data removal, rt trimming, and within subject outlier handling).

RM-ANOVA results (Accuracy, accuracy).

For effects involving the 3 level factor **Emotion** (and the interaction) **Greenhouse–Geisser (GG)–corrected p-values** are reported.

- **Main effect of face gender:** $F(1, 17) = 0.364$, $p = 0.5543$, $\eta^2 = 0.00871$ (not significant).
- **Main effect of emotion:** $F(2, 34) = 15.953$, $p_{GG} = 3.889 \times 10^{-5}$, $\varepsilon_{GG} = 0.8709$, $\eta^2 = 0.1496$ (significant).
- **Face gender × emotion interaction:** $F(2, 34) = 29.213$, $p_{GG} = 9.0 \times 10^{-8}$, $\varepsilon_{GG} = 0.9468$, $\eta^2 = 0.2964$ (significant).

Because the **interaction was significant** emotion effects were interpreted via **simple effects post hoc tests within each face-gender level**, with **Holm correction** for multiple comparison.

Post-hoc (Emotion contrasts within each face_gender; paired-samples tests, Holm-corrected).

- **Female faces**

- Angry vs Neutral: $t(17) = -6.359$, $p_{\text{Holm}} = 1.426 \times 10^{-5}$, Hedges' $g = -2.309$
- Angry vs Happy: $t(17) = -9.453$, $p_{\text{Holm}} = 1.10 \times 10^{-7}$, Hedges' $g = -2.368$
- Neutral vs Happy: $t(17) = 0.138$, $p_{\text{Holm}} = 0.8921$, Hedges' $g = 0.0416$ (not significant)

Interpretation: For **Female-face** stimuli, **Angry** trials showed **lower accuracy** than both **Neutral** and **Happy**, while Neutral and happy did not differ.

- **Male faces**

- Angry vs Neutral: $t(17) = 4.151$, $p_{\text{Holm}} = 0.001336$, Hedges' $g = 0.848$
- Angry vs Happy: $t(17) = -0.172$, $p_{\text{Holm}} = 0.8656$, Hedges' $g = -0.052$ (not significant)
- Neutral vs Happy: $t(17) = -4.430$, $p_{\text{Holm}} = 0.001101$, Hedges' $g = -0.951$

Interpretation: For **Male face** stimuli **neutral** accuracy was **lower** than both **Angry** and **Happy** whereas Angry and Happy did not differ.

Descriptive pattern (means $\pm$ SD across participants).

- Female faces: Angry **0.8473 $\pm$ 0.0533**, Neutral **0.9610 $\pm$ 0.0423**, Happy **0.9593 $\pm$ 0.0378**
- Male faces: Angry **0.9532 $\pm$ 0.0673**, Neutral **0.8895 $\pm$ 0.0792**, Happy **0.9566 $\pm$ 0.0569**

Overall these results indicate that **accuracy varies across emotions**, and critically, that **direction of the emotion effect depends on stimulus face gender**, consistent with the significant **face_gender $\times$ Emotion** interaction.

6) Discussion

The current study tested a core claim of classic face-processing accounts (e.g., the Bruce & Young framework as discussed in Ward) that **facial identity-related judgments (here: gender decisions)** and **facial expression processing** are functionally independent. In our gender-decision task, participants categorized the sex of KDEF faces while instructed to ignore expression (Happy, Angry, Neutral), using a **2 (face gender) $\times$ 3 (emotion)** within-subject design with **RT** and **accuracy** as dependent measures.

6.1 Summary of key findings relative to the hypothesis

Our primary hypothesis predicted **no RT differences across emotional conditions**, consistent with functional independence (i.e., emotion should not slow gender decisions). The RT results were broadly consistent with this prediction: **Emotion did not significantly modulate RT**, and there was **no significant face-gender $\times$ emotion interaction** in RT. Descriptively, responses tended to be slower for **female-face stimuli** than for male-face

stimuli, but the corresponding main effect was only near-threshold. Taken together, the RT pattern supports the interpretation that **expression did not reliably interfere with the speed of gender categorization** in this paradigm.

However the accuracy results qualified the “independence” conclusion. Accuracy showed a **robust emotion main effect** and critically **strong face-gender × emotion interaction**: for **female faces**, Angry trials were less accurate than Neutral/Happy, whereas for **male face**, Neutral was less accurate than Angry/Happy. This indicates that, even if **response speed** is relatively stable across emotions **decision quality** (correct categorization) can depend strongly on expression in way that differs for male versus female faces.

6.2 Interpretation

A parsimonious interpretation is that expression and identity/gender processing may be **approximately independent at the level of response latency**, yet still interact at stages that affect **evidence quality**, ambiguity resolution, or response selection—especially under more demanding stimulus conditions. This is particularly plausible given our stimulus manipulation: faces were **cropped to remove hair/ear cues**, intentionally increasing reliance on internal facial features and thereby making gender judgments more effortful and potentially more sensitive to expression-driven changes in perceived structure (e.g., mouth/eyebrow configurations).

The interaction pattern suggests the interference is not simply “Angry is always harder,” but rather that **specific expression–gender combinations** can be more confusable. This aligns with the broader motivation articulated in the design report: evolutionary/attentional accounts predict that threat-related signals (Angry faces) can capture processing resources automatically, potentially overriding strict modularity under some conditions. In our data, the strongest deviations from independence emerged in **accuracy**, consistent with the idea that expression may bias or distort feature sampling without necessarily prolonging decision time in an “until response” task.

6.3 Methodological considerations and limitations

Several factors should temper interpretation:

1. **Task timing and speed–accuracy dynamics.** The final implementation used a **wait-until-response** design to avoid missed trials, improving completeness but also allowing participants to trade speed for accuracy differently across conditions. This can reduce RT sensitivity to subtle interference while leaving accuracy effects intact.
2. **Sample characteristics.** Although the final analysis retained 18 participants, demographic heterogeneity (age range, education) and modest sample size limit generalizability and may reduce power for smaller RT effects.
3. **RT distribution and assumptions.** RT distributions were right-skewed (as expected), and normality checks suggested RT means deviated from normality, which can reduce ANOVA sensitivity unless transformations or robust models are used.

6.4 Implications and future directions

Overall the findings provide **partial support** for functional independence: **emotion did not reliably slow gender decisions**, consistent with the original RT based hypothesis. At the

same time **accuracy interaction** indicates that expression can systematically influence gender categorization outcomes, particularly under conditions of increased perceptual difficulty (cropped cues). Future work could (i) increase sample size and standardize hardware (lab-based testing), (ii) explicitly model speed accuracy tradeoff (e.g., diffusion modeling), (iii) analyze trial level data using mixed effects models, and (iv) test whether the interaction replicates with uncropped stimuli or with an explicit emotion task to probe bidirectionality.

7) Reference

- Ward, J. (2019). The Student's Guide to Cognitive Neuroscience (4th ed.). Routledge.
 - Bruce, V., & Young, A. (1986). Understanding face recognition. British Journal of Psychology, 77(3), 305–327.
 - Lundqvist, D., Flykt, A., & Öhman, A. (1998). The Karolinska Directed Emotional Faces (KDEF) [Database of standardized facial images]. Psychology Section, Department of Clinical Neuroscience, Karolinska Institutet.
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8) Endorsement of team members

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